TECHNICAL SUMMARY OF UNDERGRADUATE THESIS

# Open-Source RFIC Design and Verification of a 28-GHz Two-Stage Class-A SiGe Power Amplifier with Machine-Learning-Assisted Optimization

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| **Frequency:** 28 GHz (mmWave) | **Topology:** Two-stage Class-A PA | **Technology:** IHP SG13G2 130-nm SiGe BiCMOS Open PDK |
| **Simulation:** Qucs-S, ngspice, Xyce, Python | **Layout / EM:** KLayout, openEMS (TopMetal2 lines) | **Status:** Simulation-based; not fabricated |

## 1. Research Objective and Motivation

My undergraduate thesis investigated the design and simulation-based verification of a two-stage Class-A SiGe power amplifier (PA) centered at 28 GHz using the IHP SG13G2 130-nm SiGe BiCMOS Open PDK. The main objective was to develop the amplifier progressively from an ideal-passive circuit toward a more physically representative PDK- and EM-aware implementation, while monitoring the resulting changes in gain, impedance matching, stability, output power and efficiency.

At 28 GHz, ideal circuit elements cannot fully represent conductor loss, substrate interaction, parasitic capacitance, finite passive quality factor, distributed phase behavior and other geometry-dependent effects. The project therefore focused not only on obtaining a high-performing schematic, but also on understanding how that performance changed as increasingly realistic physical models were introduced. Machine learning was used only as a supporting layer for bounded design-space exploration, final design acceptance remained based on direct circuit simulation rather than ML prediction.

## 2. Two-Stage PA Architecture and RF Design

The PA uses a two-stage Class-A architecture consisting of a driver stage followed by a larger output stage (Figure 1). The driver provides gain and sufficient RF drive for the second stage, while the output stage provides greater current and output-power capability. Both stages use IHP SG13G2 SiGe HBTs (multiplicities $N_{x1} = 8$ and $N_{x2} = 10$) operating from a nominal 1.8-V collector supply, with independent base biases of 0.95 V and 1.04 V.

Three matching sections were required: an input matching network, an interstage network connecting two complex non-50-Ω impedances, and an output network transforming the output-stage impedance toward the external 50-Ω load. Each stage was first characterized standalone ($Z_{in,1} \approx 39.6 - j30.8\ \Omega$; $Z_{in,2} \approx 47.1 - j35.6\ \Omega$). Matching was then developed through impedance analysis, Smith-chart inspection, cascading and iterative retuning, because the impedance seen by each stage changed after integration. For example, the input L-network moved from about 59 fF / 290 pH standalone to 65 fF / 260 pH in the complete amplifier. At the interstage, a single 143-fF shunt capacitor together with the coupling capacitor was retained instead of a higher-order network, accepting a small residual reactive mismatch in exchange for lower loss, area, and tuning sensitivity.

Small-signal analysis was performed in Qucs-S using ngspice to evaluate S-parameters, impedance behavior, and stability (Rollett K, |Δ|, and μ). Large-signal input-power sweeps were performed using Xyce, with Python-based extraction of output power, saturated output power, DC power and power-added efficiency (PAE).

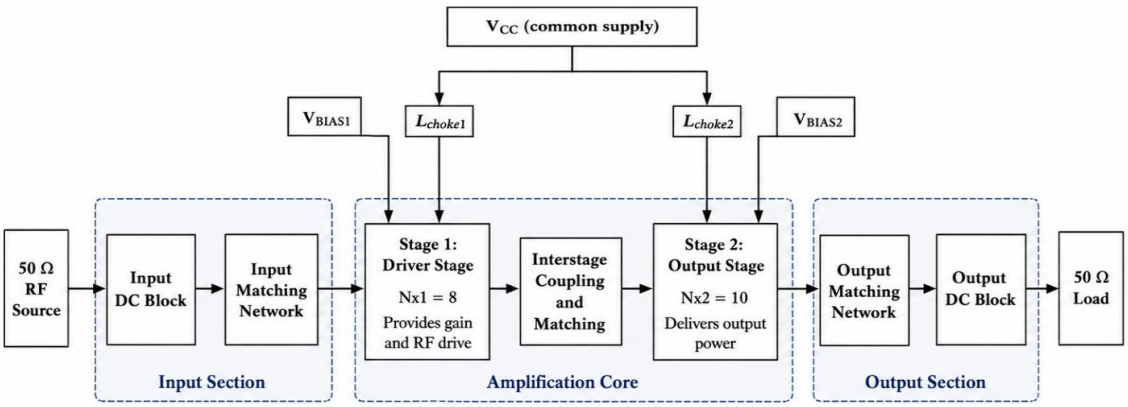



**Figure 1.** Overall architecture of the proposed 28-GHz two-stage Class-A SiGe power amplifier.

## 3. Progressive PDK and EM Realization

After the ideal-passive circuit was frozen, applicable resistors and capacitors were replaced with geometry-controlled IHP SG13G2 PDK components. Resistors were implemented using rppd structures and capacitors using RFCMIM cells, with geometry adjusted in KLayout to obtain the required nominal values (for example, a 64.2-fF RFCMIM capacitor and a 101.2-Ω rppd resistor).

The inductive elements were treated separately because ideal lumped inductors become increasingly inaccurate at mmWave frequencies. Selected inductors were implemented as TopMetal2 transmission-line structures. Their geometry was prepared in KLayout, analyzed through an openEMS-compatible electromagnetic workflow, exported as two-port S-parameter (Touchstone .s2p) models, and then reintroduced into Qucs-S (Figure 2). The EM-characterized structures covered target inductances from approximately 13 pH to 260 pH near 28 GHz. Local EM verification proved necessary rather than optional: a 140-µm line initially considered for a 30-pH element produced about 44.7 pH, whereas the accepted 90-µm line gave 30.0 pH, and the 260-pH element required an 820-µm line rather than the shorter analytical estimate.

Each EM-derived model was inserted into the PA individually rather than replacing all passives at once. This allowed the influence of conductor loss, substrate interaction, parasitic loading, finite Q, and distributed phase behavior to be observed at each checkpoint.

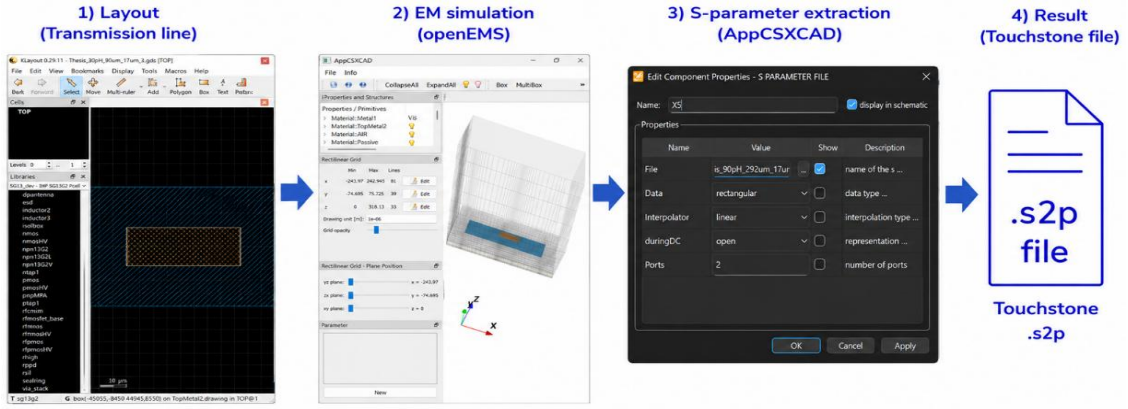



**Figure 2.** EM characterization flow for transmission-line equivalent inductors: TopMetal2 layout in KLayout, openEMS simulation, S-parameter extraction, and the resulting .s2p model used in Qucs-S.

## 4. Key Small-Signal Results

**Table 1.** Simulated small-signal performance at 28 GHz at three passive-realization checkpoints.

| Configuration (28 GHz) | $S_{21}$ (dB) | $S_{11}$ (dB) | $S_{22}$ (dB) | $S_{12}$ (dB) | K | \|Δ\| |
|---|---|---|---|---|---|---|
| Ideal-passive baseline | 22.248 | −14.8 | −32.0 | −42.6 | 5.06 | 0.0044 |
| PDK resistors + capacitors | 22.149 | −18.3 | −11.1 | −41.9 | 4.39 | 0.0325 |
| **Final PDK/EM-aware circuit** | **17.403** | **−13.3** | **−13.2** | **−45.8** | **12.0** | **0.0476** |

The ideal-passive circuit produced 22.248 dB forward gain, but this reduced to 17.403 dB (−4.845 dB) after the progressive introduction of process-specific and EM-derived passive models. The largest steps occurred when the two 30-pH lines and the 90-pH line were inserted (22.212 → 20.152 → 18.954 dB), as shown in Figure 3. The reduction should not be interpreted as a design failure. Instead, it quantifies the physical effects that the ideal schematic omitted. Despite the additional loss and parasitic loading, the final configuration retained input and output matching better than −13 dB, strong reverse isolation ($S_{12}$ = −45.8 dB) and unconditional stability by the Rollett criteria (K = 12.0, |Δ| = 0.0476).

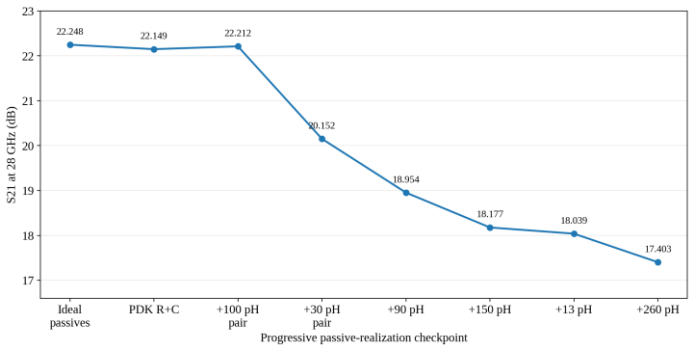



**Figure 3.** Evolution of simulated forward gain ($S_{21}$ at 28 GHz) as PDK and EM-derived passive models were introduced progressively.

## 5. Large-Signal and Design-Space Evaluation

Large-signal analysis was performed using Xyce through Qucs-S with Python-based post-processing. Input-power sweeps were used to extract output power, saturated output power ($P_{sat}$), DC power consumption, gain-compression behavior, and peak PAE. Under an earlier, stricter output-matching comparison ($S_{22} \leq -10$ dB), the strongest directly simulated design achieved approximately 17.7 dB forward gain, 12.1 dBm saturated output power and 24.5% peak PAE.

## 6. ML-Assisted Design-Space Exploration and Design Tool

A structured simulation dataset was also generated by varying five circuit quantities: the common collector supply voltage, the two stage-bias voltages, and the two collector-choke inductances, while the topology, transistor sizes, and matching networks were held fixed. The complete ML-ready dataset contained 120 records, with 92 simulator-derived records used as formal modeling evidence and 28 imputed records retained only for supplemental use. Target-specific surrogate models were then used to screen a bounded candidate space, but shortlisted designs were always returned to Qucs-S for independent verification.

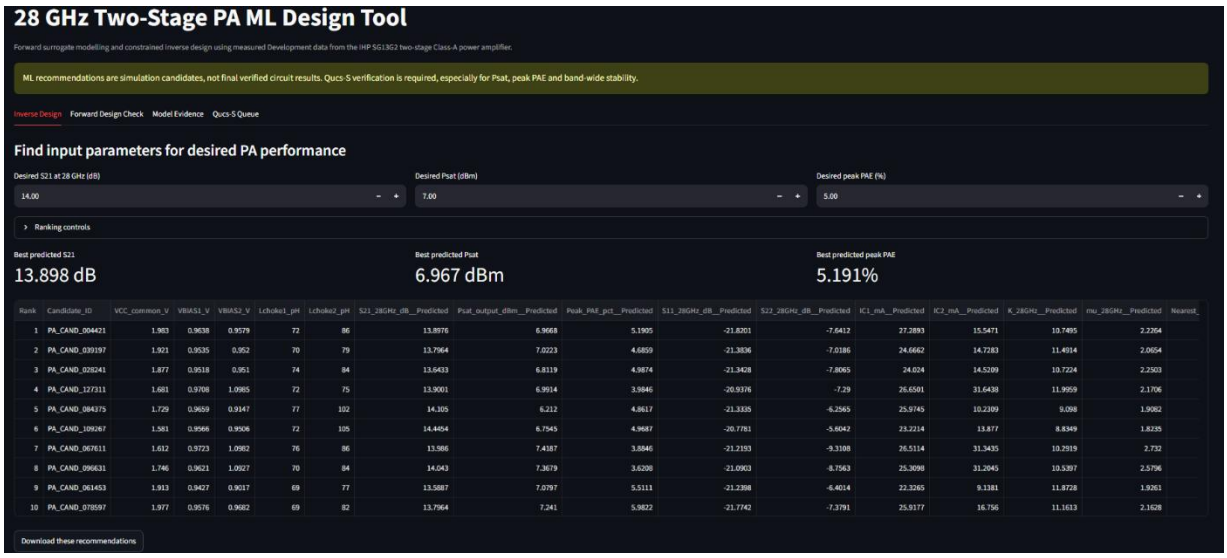



**Figure 4.** Streamlit-based 28-GHz PA design-assistance interface for target-driven candidate recommendation using the frozen surrogate models.

To make the surrogate models usable as a practical design-assistance tool, I developed a Streamlit-based interface that accepts desired PA performance targets and returns ranked candidate circuit settings from the screened design library. As shown in Figure 4, the interface reports the predicted gain, saturated output power, peak PAE and associated circuit and constraint quantities for the recommended candidates. The recommendations are treated only as simulation candidates rather than verified circuit designs and selected candidates must still be evaluated through direct Qucs-S simulation before acceptance.

## 7. Scope and Current Limitations

The completed work is a simulation-based, schematic-level PDK/EM-aware RFIC study. The physical-realization scope included geometry-controlled PDK resistors and capacitors together with selected EM-characterized inductive structures. Full-chip layout, complete interconnect extraction, final DRC/LVS sign-off, fabrication, packaging and measured-silicon characterization were outside the completed scope. In addition, some simulator-compatible substitutions were required in the large-signal netlist, so the large-signal results and the final small-signal EM-aware checkpoint do not use identical passive implementations. Accordingly, the reported results should be interpreted as progressively process-aware simulation results rather than fabricated or measured-silicon performance.

## 8. Research Direction

This thesis gave me practical experience in mmWave PA architecture, transistor biasing, impedance matching, S-parameter and stability analysis, PDK-aware passive implementation, EM modeling, and simulation-based verification. As a post-thesis extension, I am currently extending the work toward a full-custom layout implementation of the PA using the IHP SG13G2 PDK, including physical placement and routing, layout refinement and DRC/LVS verification, with the goal of producing a clean final GDSII representation.

Beyond this ongoing layout work, I would like to develop further experience in parasitic extraction, post-layout RF optimization, tapeout-oriented verification, fabrication and high-frequency measurement. My broader goal is to continue working in RF/mmWave integrated-circuit design and gain experience across the complete path from schematic design and EM-aware modeling to verified layout and measured hardware.